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import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression

from sklearn.preprocessing import PolynomialFeatures

from sklearn.metrics import mean_squared_error, r2_score


# Load the dataset

df = pd.read_csv("data/auto-mpg.csv")


print("First 5 rows:")

print(df.head())

print("\nDataset shape:")

print(df.shape)


# Preprocess the data

df.dropna(inplace=True)

X = df[["displacement"]]

y = df["mpg"]


# Split the dataset

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42 )


# Linear Regression

linear_model = LinearRegression()

linear_model.fit(X_train, y_train)

y_linear_pred = linear_model.predict(X_test)

linear_mse = mean_squared_error(y_test, y_linear_pred)

linear_r2 = r2_score(y_test, y_linear_pred)

print("\n===== LINEAR REGRESSION =====")

print("MSE :", linear_mse)

print("R2 Score :", linear_r2)


# Polynomial Regression

degrees = [2, 3, 4, 5]

results = []

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for degree in degrees:
    poly = PolynomialFeatures(degree=degree)
    X_train_poly = poly.fit_transform(X_train)
    X_test_poly = poly.transform(X_test)
    poly_model = LinearRegression()
    poly_model.fit(X_train_poly, y_train)
    y_poly_pred = poly_model.predict(X_test_poly)
    mse = mean_squared_error(y_test, y_poly_pred)
    r2 = r2_score(y_test, y_poly_pred)
    results.append([degree, mse, r2])
    print("\n===== POLYNOMIAL REGRESSION =====")
    print("Degree :", degree)
    print("MSE :", mse)
    print("R2 Score :", r2)

```

Model Comparison

```

print("\n===== MODEL COMPARISON =====")
print("Linear Regression")
print("MSE :", linear_mse)
print("R2 Score :", linear_r2)

```

for result in results:

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    print(
        "Polynomial Degree", result[0],
        "MSE :", result[1],
        "R2 Score :", result[2] )

```

Visualization

for degree in degrees:

```

    poly = PolynomialFeatures(degree=degree)
    X_poly = poly.fit_transform(X)
    poly_model = LinearRegression()
    poly_model.fit(X_poly, y)
    X_plot = np.linspace(
        X["displacement"].min(),
        X["displacement"].max(),
        300 ).reshape(-1, 1)
    X_plot_poly = poly.transform(X_plot)
    y_plot = poly_model.predict(X_plot_poly)
    plt.scatter(
        X, y, color="blue", label="Actual Data" )

```

```
plt.plot(
    X_plot, y_plot, color="red", linewidth=2, label="Polynomial Regression" )

plt.xlabel("Engine Displacement")
plt.ylabel("Miles Per Gallon (MPG)")
plt.title("Polynomial Regression: Displacement vs MPG")

plt.legend()

plt.grid()

plt.show()
```

First 5 rows:

	mpg	cylinders	displacement	horsepower	weight	acceleration	model year	origin	car name
0	18.0	8	307.0	130	3504	12.0	70	1	chevrolet chevelle malibu
1	15.0	8	350.0	165	3693	11.5	70	1	buick skylark 320
2	18.0	8	318.0	150	3436	11.0	70	1	plymouth satellite
3	16.0	8	304.0	150	3433	12.0	70	1	amc rebel sst
4	17.0	8	302.0	140	3449	10.5	70	1	ford torino

Dataset shape:
(398, 9)

===== LINEAR REGRESSION =====

MSE : 18.102543998358946

R2 Score : 0.6633114869465596

===== POLYNOMIAL REGRESSION =====

Degree : 2

MSE : 15.10735356108076

R2 Score : 0.7190189176110284

===== POLYNOMIAL REGRESSION =====

Degree : 3

MSE : 16.369649051540456

R2 Score : 0.6955415327884606

===== POLYNOMIAL REGRESSION =====

Degree : 4

MSE : 18.39879031020308

R2 Score : 0.6578016132933643

===== POLYNOMIAL REGRESSION =====

Degree : 5

MSE : 20.998145709208522

R2 Score : 0.6094563031387301

===== MODEL COMPARISON =====

Linear Regression

MSE : 18.102543998358946

R2 Score : 0.6633114869465596

Polynomial Degree 2 MSE : 15.10735356108076 R2 Score : 0.7190189176110284

Polynomial Degree 3 MSE : 16.369649051540456 R2 Score : 0.6955415327884606

Polynomial Degree 4 MSE : 18.39879031020308 R2 Score : 0.6578016132933643

Polynomial Degree 5 MSE : 20.998145709208522 R2 Score : 0.6094563031387301

